



e u r e x circular 178/05

Date: Frankfurt, September 26, 2005
Recipients: All Eurex members and vendors, CCP members
Authorized by: Peter Reitz



High Priority

Trading Hours Extension on Eurex Exchanges (THX): **Detailed Functional Specifications**

Related Eurex Circular: 116/05

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Content may be most important for:



All departments

Attachment:

- THX - Eurex Exchanges Trading Schedule "new"

Summary:

As announced in Eurex circular 116/05 on July 7, 2005, Eurex is planning to extend trading hours in certain Eurex benchmark products to 22:00 CET (15:00 CST) from **November 21, 2005** on.

The present circular provides additional information to support Eurex market participants in their activities regarding a successful start of the trading hours extension (THX).

The circular covers the following subjects:

- **Products in Scope**
- **Trading Schedule**
- **Eurex Settlement Price Procedure**
- **New Message Timeline**
- **Availability of qualified Back Office Personnel**
- **Simulation – Production-like week**

If you have any questions or request any further information, please contact the Customer Support team at telephone +49-69-211-1 17 00 or fax +49-69-211-1 17 01.



Trading Hours Extension on Eurex Exchanges (THX): **Detailed Functional Specifications**

Products in Scope

According to the current schedule, trading hours extension (THX) will apply to the following Eurex benchmark products:

- Interest Rate Products
 - Euro Buxl® Futures
 - Euro Bund Futures
 - Euro Bobl Futures
 - Euro Schatz Futures
- Equity Index Products
 - All equity index futures
(with the exception of SMI® Futures, SMIM® Futures and Volatility Futures)
 - Futures on Shares of EUR EXTfs

Trading Schedule

With THX, trading will close at 22:00 CET for all products in scope. The post-trading period is scheduled for 22:00 to 22:30 CET. Subsequently, all Eurex batch information will be provided approximately two hours later than currently. The “End-of-Day” Theoretical Price File will be expected for approximately 23:30 CET (times may differ each business day).

With THX, Eurex will also extend its current OTC block trading times. For more detailed information, please refer to the attached trading schedule.

Eurex Daily Settlement Price Procedure

The determination of daily settlement prices will be described in the Clearing Conditions for Eurex Clearing AG. In general, the following procedure applies:

Interest rate products (in scope): Determination of the daily settlement prices will take place at approximately 17:15 CET, based on the volume-weighted average price of all trades between 17:14 CET and 17:15 CET.

Equity index products in scope: Determination of the daily settlement prices will take place at approximately 17:30 CET, based on the volume-weighted average price of all trades between 17:29 CET and 17:30 CET.

The determined settlement prices will be published via the Eurex® system. Trading will continue without interruption until 22:00 CET. Please note that the closing auction at 22:00 CET will include market on close orders whereby the closing price will not be relevant for the settlement price.

For detailed information regarding THX for products both in scope and out of scope, please refer to the trading schedule attached.

New Message Timeline

After all settlement prices have been approved by Eurex Clearing AG, an “intra-day” Theoretical Price File will be delivered to the member MISSES at approximately 19:00 CET (exception: MISSES connected via Internet) and in addition published via the Eurex website also at approximately 19:00 CET. Please note that the distribution via MISSES is closely linked to the Eurex network upgrade and therefore the required bandwidths should be available. If this is not the case, distribution of the “intra-day” Theoretical Price File might affect the individual system performance.

Please note that in general the “intra-day”¹ and the “end-of-day” Theoretical Price Files - except the file name - will not differ. The “intra-day” Theoretical Price File will have the name:

- **THEA_INTRYMMDDnn.DAT** (published via MISS)
(nn = 00 for production and 01 for simulation)
- **Tlyyyymmdd.dat** (published via Internet)

According to existing procedures, the “end-of-day” Theoretical Price File is also published via MISS and Internet. However, the new expected delivery time of the file is approximately two hours later.

Internet download² will be possible twice a business day at approximately 19:00 CET and during the batch at approximately 23:30 CET under the existing path:

www.eurexchange.com > Clearing & Settlement > Market Information > Theoreticals > Eurex Theoretical Price Data

With the implementation of THX, the existing Eurex message timeline in general will be shifted by approximately two hours. The following table provides an overview on batch information times after THX has been introduced. Please note that these times may not be the same on every business day and therefore cannot be guaranteed.

- Message -	- Current -	- Planned -
Eurex® System Status Batch	approx. 20:30 CET	approx. 22:30 CET
End of Assignment Message	approx. 20 minutes after System Status Batch	20 minutes after System Status Batch
Theo Price File/ Margin Parameter File + Flex Options Theo File	approx. 21:30 - 22:00 CET	approx. 23:30 - 00:00 CET*
Notification/Allocation on exercise days (Fixed Income Futures)	Series maintenance approx. 30 mins. after System Status Batch	Series maintenance approx. 30 mins. after System Status Batch
End of stream/ End of Batch	approx. 22:30 CET	approx 00:30 CET
Eurex Reports	untill 06:00 CET t+1	untill 06:00 CET t+1
Eurex US Theo File	approx. 22:00 CET	approx. 00:00 CET
Eurex Bonds Theo File	approx. 22:00 CET	approx. 00:00 CET
THEA File (Equity)	approx. 22:10 - 22:20 CET	approx. 23:30 - 23:40 CET
CCP - Reports	continuous sending of reports	continuous sending of reports

* Flex Options theo prices approx. at 00:00 CET

¹ Please note, that the Flex Option Theo Price File will only be sent during the Eurex batch at approx. 00:00 CET and **not** “intra-day”

² According to existing procedures download of the file via Internet will be charged with EUR 500 per month according to the current pricing. To apply for a production user ID and password, please register under: **www.eurexchange.com > Clearing & Settlement > Market Information > File Service > Open an Account**

Availability of qualified Back Office Personnel

During the business day of Eurex Clearing AG at least one sufficiently qualified member of staff must be present and available by both telephone and fax at any time for the orderly fulfillment of clearing obligations in the back office (For further details see "Exchange Rules for Eurex Deutschland and Eurex Zürich" (Chapter 3.4.1 Back Office Personnel and "Clearing Conditions for Eurex Clearing AG", Number 1.1.2 (4 f) of Chapter III, IV and V). The business day of Eurex Clearing AG covers the pre-trading, trading and post-trading phases of the connected trading systems.

Simulation – Production-like week

From **October 17 to October 20, 2005**, a production-like week is scheduled in order to allow market participants to test THX in a production-like environment. An intraday Eurex Theoretical Price File will be published on the Eurex website on a daily basis at approximately 20:00 CET. The file can be downloaded under the path:

**www.eurexexchange.com > Clearing & Settlement > Market Information > Theoreticals >
Eurex Theoretical Price Data > Eurex THX Simulation Theoretical Price Data**

Further details about the simulation week will be announced in the upcoming weeks.

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Frankfurt, September 26, 2005

Product/ Group As of November 21 2005	Pre- Trading	Trading (PO+Trad)	Post- Trading- Full	OTC Block Trading	Closing Auction/	Daily Settlement price	Last Trading day	
							trading until	Notify / Exercise until
<u>Interest Rate Derivatives</u>								
CONF Futures	07:30-08:30	08:30-17:00	17:00-20:00	08:30-17:00	17:00	17:00	12:30	20:00
Euro-Bobl Future	07:30-08:00	08:00-22:00 ¹	22:00-22:30	08:00-22:00	22:00	17:15	12:30	22:30
Euro-Bund Future	07:30-08:00	08:00-22:00 ¹	22:00-22:30	08:00-22:00	22:00	17:15	12:30	22:30
Euro-Buxl [®] Future	07:30-08:00	08:00-22:00 ¹	22:00-22:30	08:00-22:00	22:00	17:15	12:30	22:30
Euro-Schatz Future	07:30-08:00	08:00-22:00 ¹	22:00-22:30	08:00-22:00	22:00	17:15	12:30	22:30
One-Month EONIA Future	07:30-08:00	08:00-19:00	19:00-20:00	08:00-19:00	-	17:15	19:00 ²	-
Options on Euro-Bobl Future	07:30-08:00	08:00-19:00	19:00-20:00	08:00-19:00	-	17:15	17:15	18:00
Options on Euro-Bund Future	07:30-08:00	08:00-19:00	19:00-20:00	08:00-19:00	-	17:15	17:15	18:00
Options on Euro-Schatz Future	07:30-08:00	08:00-19:00	19:00-20:00	08:00-19:00	-	17:15	17:15	18:00
Options on Three-Month	07:30-08:00	08:00-19:00	19:00-20:00	08:00-19:00	-	17:15	11:00	20:00
Three-Month EURIBOR	07:30-08:00	08:00-19:00	19:00-20:00	08:00-19:00	-	17:15	11:00	-
<u>Equity Derivatives</u>								
Equity Options on Dutch Shares	07:30-08:53	08:53-17:30	17:30-20:00	09:00-19:00	-	17:30	17:30	20:00

¹ According to the current exchange rules Eurex reserves the right to adapt trading hours e.g. in case of a US holiday.

² Final Settlement Price is scheduled for approx. 19:00 CET (depending on information to be received from European Central Bank)

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Equity Options on French Shares	07:30-08:54	08:54-17:30	17:30-20:00	09:00-19:00	-	17:30	17:30	20:00
Equity Options on German Shares	07:30-08:51	08:51-17:30	17:30-20:00	09:00-19:00	-	17:30	17:30	20:00
Equity Options on Italian Shares	07:30-08:50	08:50-17:30	17:30-20:00	09:00-19:00	-	17:30	17:30	20:00
Equity Options on Scandinavian Shares	07:30-08:52	08:52-17:30	17:30-20:00	09:00-19:00	-	17:30	17:30	20:00
Equity Options on Swiss Shares	07:30-08:52	08:52-17:30	17:30-20:00	09:00-19:00	-	17:30	17:30	20:00
Equity Options on US Shares	07:30-08:53	08:53-17:30	17:30-20:00	09:00-19:00	-	17:30	17:30	20:00
<u>Equity Index Derivatives</u>								
DAX [®] Futures	07:30-08:50	08:50-22:00³	22:00-22:30	09:00-22:00	22:00	17:30	13:00	
DAX [®] Options	07:30-08:50	08:50-17:30	17:30-20:30	09:00-19:00		17:30	13:00	21:00
Dow Jones EURO STOXX [®] 50 Future	07:30-08:50	08:50-22:00³	22:00-22:30	09:00-22:00	22:00	17:30	12:00	
Dow Jones EURO STOXX [®] 50 Options	07:30-08:50	08:50-17:30	17:30-20:30	09:00-19:00		17:30	12:00	21:00

³ According to the current exchange rules Eurex reserves the right to adapt trading hours e.g. in case of a US holiday.

Product/ Group As of November 21 2005	Pre- Trading	Trading (PO+Trad)	Post- Trading- Full	OTC Block Trading	Closing Auction/	Daily Settlement price	Last Trading day	
							trading until	Notify / Exercise until
Dow Jones Global Titans Index Futures	07:30-08:50	08:50-22:00⁴	22:00-22:30	09:00-22:00	22:00	17:30	17:00	
Dow Jones Global Titans Index Options	07:30-08:50	08:50-17:30	17:30-20:30	09:00-19:00		17:30	17:00	21:00
Dow Jones Italy Titans Index Futures	07:30-08:50	08:50-22:00⁴	22:00-22:30	09:00-22:00	22:00	17:30	09:10	
Dow Jones Italy Titans Index Options	07:30-08:50	08:50-17:30	17:30-20:30	09:00-19:00		17:30	09:10	21:00
Dow Jones STOXX [®] Futures	07:30-08:50	08:50-22:00⁴	22:00-22:30	09:00-22:00	22:00	17:30	12:00	
Dow Jones STOXX [®] Options	07:30-08:50	08:50-17:30	17:30-20:30	09:00-19:00		17:30	12:00	21:00
Dow Jones STOXX [®] Sector Index Futures	07:30-09:00	09:00-22:00⁴	22:00-22:30	09:00-22:00	22:00	17:30	12:00	
Dow Jones STOXX [®] Sector Index Options	07:30-09:00	09:00-17:30	17:30-20:30	09:00-19:00		17:30	12:00	21:00
OMXH25 Futures	07:30-08:50	08:50-22:00⁴	22:00-22:30	09:00-22:00	22:00	17:30	17:30	
OMXH25 Options	07:30-08:50	08:50-17:30	17:30-20:30	09:00-19:00		17:30	17:30	21:00
SMI [®] Futures	07:30-08:50	08:50-17:30	17:30-20:30	09:00-19:00	17:30	17:30	17:30	

⁴ According to the current exchange rules Eurex reserves the right to adapt trading hours e.g. in case of a US holiday.

Product/ Group As of November 21 2005	Pre- Trading	Trading (PO+Trad)	Post- Trading- Full	OTC Block Trading	Closing Auction/	Daily Settlement price	Last Trading day	
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SMI [®] Options	07:30-08:50	08:50-17:20	17:20-20:30	09:00-19:00		17:20	17:20	21:00
TecDAX Futures	07:30-08:50	08:50-22:00⁵	22:00-22:30	09:00-22:00	22:00	17:30	13:00	
TecDAX Options	07:30-08:50	08:50-17:30	17:30-20:30	09:00-19:00		17:30	13:00	21:00
<u>Exchange Traded Funds</u>								
<u>Derivatives</u>								
Futures on shares of CHF EXTFs	07:30-08:51	08:51-17:20	17:20- 20:30	-		17:20	17:20	
Futures on shares of EUR EXTFs	07:30-08:51	08:51-22:00⁵	22:00-22:30	-		17:30	17:30	
Options on shares of CHF EXTFs	07:30-08:51	08:51-17:20	17:20-20:00	09:00-19:00		17:20	17:20	20:00
Options on shares of EUR EXTFs	07:30-08:51	08:51:17:30	17:30-20:00	09:00-19:00		17:30	17:30	20:00
<u>Vola Futures</u>								
DAX [®]	07:30-08:50	08:51:17:30	17:30-20:30	09:00-19:00	17:30	17:30	13:00	
ES50	07:30-08:50	08:51:17:30	17:30-20:30	09:00-19:00	17:30	17:30	12:00	
SMI [®]	07:30-08:50	08:51:17:20	17:20-20:30	09:00-19:00	17:20	17:20	10:00	

⁵ According to the current exchange rules Eurex reserves the right to adapt trading hours e.g. in case of a US holiday.